

Executive Summary: Full-stack client acquisition and social publishing platform for Klyperix. Integrates real-time lead generation (Google Maps, YouTube Creators, Reddit Hiring Posts, Live Web Scraper), AI outreach writing (Groq Llama 3.3 70B), multi-channel outreach (WhatsApp with 40-msg/day anti-ban & Gmail OAuth 2.0), and a 1-Click Multi-Platform Social Media Publishing Studio.

1. WHAT WE CHANGED (Major Modifications)

- **AI Social Output Normalization:** Standardized keys between backend AI (Groq Llama 3.3) and frontend so all 5 platforms (Instagram, LinkedIn, YouTube Shorts, X, Facebook) reliably display native headlines, captions, takeaways, and hashtags.
- **1-Click Social Publishing Flow:** Replaced static status badges with an active Multi-Platform Launchpad modal. Clicking '1-Click Publish' saves the post, generates verified web intents, and opens 1-click dispatch controls with auto-clipboard integration.
- **Modernized Web Intent Endpoints:** Upgraded sharing endpoints to modern standards (x.com/intent/post, pre-filled LinkedIn feed composer, Facebook sharer, YouTube Studio upload, Instagram Creator Studio).
- **Real Data Pipelines replacing Mock Stubs:** Refactored search pipelines to use real production services (OpenStreetMap / Google Places API, YouTube Data API v3, Reddit Script API, native Web Contact Crawler) instead of simulated demo leads.
- **TypeScript Strict Compilation:** Resolved all typing errors across server and client codebase. Both 'npm run build:server' and 'npm run build:client' compile cleanly with zero errors (Exit Code 0).

2. WHAT WE ADDED (New Features & Capabilities)

- **Social Media Client Credentials UI (Settings Tab):** Added dedicated fields in Settings for LinkedIn Client ID & Secret, X (Twitter) API Key & Secret, and Meta (Instagram & Facebook) App ID & Token.
- **AI Content Studio & 1-Click Multi-Publish (Tab #6):** Dedicated studio for Reels, Videos, Images, Carousels, and Posts. Supports dual brand modes ('Klyperix Production' & 'Klyperix Gems & Jewels') with single-click AI generation across 5 platforms.
- **Interactive Multi-Platform Launchpad Modal:** Dedicated dispatch modal for Instagram, LinkedIn, YouTube, X, and Facebook with live text preview, one-click copy buttons, and direct 'Open & Post' buttons.
- **In-Tab Direct Action Launcher:** Within each preview tab under '2. AI Native Formats', added a direct 'Post on [Platform] Now' button that copies formatted content and opens the official composer.
- **Content Pipeline Management (Launch & Delete):** Added 'Launch / Re-post' button to relaunch any past post, and 'Delete' button to permanently purge unwanted pipeline records.
- **WhatsApp Native Web Engine with Anti-Ban Guard:** Direct Baileys QR web socket pairing with a mandatory 40-message daily rate limit, automated pacing delays (12-25s), and auto-pause protection.
- **Official Gmail Sign-In via Google OAuth 2.0:** One-click OAuth 2.0 login under client's personal/workspace Google account, storing offline refresh tokens to send verified emails directly.

3. WHAT WE REMOVED (And Why)

- **Gmail App Password (SMTP) & Unneeded Signature Cards:** Removed: Confusing manual SMTP passwords and extra signature cards removed to keep Settings clean, in favor of 1-click Google Sign-In and Social Credentials.
- **Fabricated Freelance Platforms (Upwork, Freelancer, Fiverr):** Removed: Scraping freelance sites without enterprise API access violates their Terms of Service and guarantees IP/account termination. Replaced with 100% legal Google Maps and Reddit hiring posts.
- **Unauthorized Private DM Bots (Instagram / X / LinkedIn):** Removed: Private DM automation bots risk permanent ban of client social handles. Replaced with 100% TOS-compliant Web Intent Launchers and verified contact scrapers.
- **Static Mock Indicators & Hardcoded Statuses:** Removed: Replaced all fake 'connected' indicators with live token/API health verification.

4. APIS & CLIENT CREDENTIALS SPECIFICATION

The platform is built with a **Zero-Cost Baseline**: WhatsApp outreach, OpenStreetMap lead search, and 1-Click Social Posting work out-of-the-box with zero API keys. Below is the complete credential reference for optional upgrades and AI engines.

Service	Credential / Env Key	Where to Obtain	Status / Necessity	Purpose & Function
Groq AI (Llama 3.3 70B)	GROQ_API_KEY	console.groq.com/keys	Pre-Configured	Writes AI outreach pitches, website audits, and 5 native social copies.
Google Sign-In (Gmail OAuth 2.0)	GOOGLE_CLIENT_ID GOOGLE_CLIENT_SECRET	console.cloud.google.com (Enable Gmail API)	Recommended (One-time setup)	Enables 1-click 'Sign in with Google' to send real emails from client's Gmail address.
Google Maps (Places API)	GOOGLE_PLACES_API_KEY	console.cloud.google.com (Enable Places API)	Optional Upgrade (Free OSM Active)	Upgrades local business search with verified Google ratings, reviews, and places data.
YouTube Search (Data API v3)	YOUTUBE_API_KEY	console.cloud.google.com (Enable YouTube v3)	Optional (For Creator Niche)	Discovers YouTube channels, creator subscriber counts, video volume, and topics.
Reddit Hiring (Script App)	REDDIT_API_KEY (id:secret)	reddit.com/prefs/apps (Create 'script' app)	Optional (For Job Posts)	Searches active hiring subreddits (r/forhire, r/freelance_forhire) for paid gigs.
WhatsApp Web (Baileys Engine)	None Required (Native QR Link)	Connections Hub -> Click 'Pair WhatsApp' & scan	Zero Keys	Direct phone link like WhatsApp Web. Includes 40-msg/day anti-ban rate limiting.
Social Publishing (Current Mode)	None Required (Web Intent Mode)	1-Click Launchpad directly launches official web composers	Zero Keys	100% compliant, zero ban risk. Auto-copies captions & hashtags to clipboard.
Social Auto-Post (Optional Future API)	LinkedIn: Client ID/Secret X: API Key + Secret Meta: App ID + Token	LinkedIn Devs / X Dev Portal / Meta Developers	Optional (Needs App Review)	Required ONLY if client wants background automated posting without browser tabs.

5. CLIENT DELIVERY CHECKLIST & INSTRUCTIONS

- **1. Running the Platform:** Run npm run dev from project root. Access frontend on http://localhost:5173 and backend on http://localhost:3001.
- **2. Connecting WhatsApp:** Go to **Connections** tab -> WhatsApp card -> Click **Pair WhatsApp** -> Scan QR code with phone camera / WhatsApp Linked Devices (capped at 40 msgs/day for safety).
- **3. Connecting Gmail:** Add GOOGLE_CLIENT_ID and GOOGLE_CLIENT_SECRET in server/.env -> Click **Sign in with Google** -> Grant Gmail permissions.
- **4. Launching Social Posts:** Go to **Content Studio** (Tab #6) -> Select Content Type -> Enter core idea -> Click **Generate Multi-Platform AI Copy** -> Click **1-Click Publish to All** to launch across Instagram, LinkedIn, YouTube, X, and Facebook!